

VAIBHAV GARG

Data Engineer | Software Engineer | Python | PySpark | AWS | Snowflake | SQL | ETL Pipelines | Power BI
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Post-Study Work Visa (Subclass 485) — 4 years remaining | No sponsorship required | Available Immediately
Open to Sydney · Melbourne · Brisbane · Perth · Adelaide

PROFESSIONAL SUMMARY

Results-driven Data Engineer and Software Engineer with 3 years of production experience and a Master of Artificial Intelligence and Machine Learning from the University of Adelaide, Australia (GPA 6.6/7.0, 2025). Specialises in building scalable ETL/ELT pipelines, cloud-native data architectures, and distributed data processing systems using Python, PySpark, SQL, AWS, and Snowflake. Delivered consistent, measurable outcomes across enterprise-scale environments: 30% reduction in pipeline execution time, 5+ TB of operational data processed daily with zero SLA breaches, 95%+ elimination of production data errors, and a zero-incident multi-cloud migration. Complemented by applied AI/ML expertise — NLP, computer vision, predictive modelling — and a strong software engineering foundation in RESTful APIs, test automation, CI/CD, and code quality. Fully work-authorised in Australia on a Post-Study Work Visa (Subclass 485) with 4 years remaining. No sponsorship required. Available to start immediately.

CORE SKILLS

Languages & Query: Python (advanced), SQL (advanced), PySpark, Java (basic), Shell Scripting

Data Engineering: ETL/ELT Pipeline Design, Apache Spark, PySpark, Hadoop, Hive, Data Ingestion, Transformation, Data Quality, Data Lineage, Airflow, dbt

AWS: S3, Glue, Lambda, EC2, Redshift, IAM, CloudWatch, EMR, Athena, DMS, API Gateway, SNS, SQS, DynamoDB

Cloud & Warehousing: Snowflake, GCP, Azure (basics), Data Lakes, Dimensional Modelling, Medallion Architecture, Multi-cloud

Python Ecosystem: Pandas, NumPy, Scikit-learn, SQLAlchemy, FastAPI, Flask, pytest, Matplotlib, Seaborn

ML & AI: Scikit-learn, Regression, Classification, Clustering (K-Means), Time-Series, NLP, Computer Vision (YOLOv8, OpenCV), Model Deployment

Software Engineering: RESTful API Design, Microservices, OOP, Secure Coding, Code Review, Performance Optimisation, Spring Boot

DevOps & CI/CD: Git, GitHub Actions, Docker, AWS CodePipeline, pytest — unit & integration testing

BI & Monitoring: Power BI, CloudWatch, Structured Logging, SLA Tracking, Root-Cause Analysis

Databases: Snowflake, Redshift, PostgreSQL, MySQL, DynamoDB, SQLite — schema design, migrations, query optimisation

PROFESSIONAL EXPERIENCE

Software Engineer | Impetus Technologies, India

Aug 2021 – Dec 2023

Python · PySpark · SQL · AWS (S3, Glue, Lambda, EC2, Redshift, CloudWatch, IAM) · Snowflake · GCP · REST APIs · Git · Agile

- **Achieved 30% faster pipeline execution** — re-architected distributed PySpark ETL pipelines on AWS Glue processing 5+ TB daily; redesigned partition strategy, eliminated shuffle-heavy joins, and parallelised sequential jobs with zero SLA breaches post-deployment.
- **Reduced production data errors by 95%+** — built Python data-quality validation frameworks enforcing schema integrity, null/duplicate detection, and referential consistency at every AWS Glue ingestion stage; zero downstream rework requests after implementation.
- **Delivered zero-incident cloud migration** — migrated legacy on-premise databases to AWS (S3, Glue, Lambda) and Snowflake; designed dimensional schemas, built Python reconciliation scripts, onboarded BI teams within 8 weeks.
- **Processed thousands of daily API requests with zero security incidents** — developed RESTful APIs (Spring Boot) with role-based authentication, input validation, and structured error handling powering an ML loan eligibility scoring engine.
- **Cut pipeline failure detection from hours to under 15 minutes** — configured CloudWatch custom metrics, alarms, and log groups; maintained runbooks and SOPs in Git for incident response and knowledge transfer.
- **Delivered ML solutions across Retail, Finance, and Operations verticals** — built classification, regression, time-series, and clustering models using Python and Scikit-learn for international cross-functional client teams.
- **Maintained delivery quality across Agile sprints** — participated in code reviews, managed feature branches, automated test pipelines with pytest, and prepared weekly leadership reports for international stakeholders.

Console Operator (Responsible Person) | OTR Group — Viva Energy, Adelaide

Apr 2024 – Present

Power BI · Operational Analytics · Incident Management · SLA Compliance

- **Applied Power BI in a live production environment** — monitored real-time store KPIs and operational metrics; identified anomalies, applied structured resolution workflows, and maintained SLA compliance alongside postgraduate study.
- **Managed operational incidents under pressure** — prioritised competing issues, applied SOPs, and maintained compliance standards; demonstrates accountability and structured problem-solving in high-stakes environments.

KEY PROJECTS

Pelmorex — Multi-Cloud Data Pipeline Migration | PySpark · AWS · Snowflake · GCP · Python | Proprietary client project

- **Delivered analytics data 2x faster with 30% reduction in manual operations** — migrated North American weather data platform to multi-cloud AWS + Snowflake + GCP; re-architected PySpark partition strategy, built Python orchestration scripts with zero SLA misses post-deployment.

ATM Transaction Analytics Pipeline | PySpark · SQL · Parquet · Docker · GitHub Actions · pytest |

github.com/vaibhavgarg003

- **12/12 pytest tests passing, full CI/CD pipeline** — production PySpark pipeline processing 10,000 ATM transactions; time-bucket classification, high-value flagging, Parquet partitioned output, Dockerised with GitHub Actions automation.

Football Tactical Formation Detection | *YOLOv8 · DeepSORT · OpenCV · K-Means · Python* | github.com/vaibhavgarg003

- **Validated against SoccerNet-v2** — computer vision pipeline detecting football formations; YOLOv8 detection, HSV colour segmentation, DeepSORT tracking, K-Means classification across a full 90-minute match.

ETL Pipeline — Country Data Engineering | *Python · Pandas · SQLAlchemy · Docker · pytest* | github.com/vaibhavgarg003

- **15/15 tests passing, 195 countries, live REST API** — production ETL pipeline; extract, transform with Pandas, load into SQLite via SQLAlchemy; Dockerised, structured logging, secrets management.

EDUCATION

Master of Artificial Intelligence & Machine Learning — University of Adelaide, Australia Jan 2024 – Dec 2025

GPA: 6.6/7.0 | *Adelaide Graduate Award (Nov 2025) — leadership, professional excellence, cross-cultural collaboration*

- Coursework: Deep learning, NLP, computer vision, reinforcement learning, time-series, ML deployment, distributed data systems.

Bachelor of Computer Science (B.E./B.Tech) — Rajasthan Technical University

2017 – 2021

80% aggregate

CERTIFICATIONS

Advanced Microsoft Power BI — Edureka, Mar 2026 | **AWS Fundamentals Specialization** — Amazon Web Services, Sep 2020

Python for Everybody Specialization — University of Michigan, Jul 2020 | **AWS Solutions Architect Associate** — Self-studying, Target Q2 2026

PROFESSIONAL REFEREES

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